

A gentle introduction to Bayesian statistics and probabilistic modeling

Part I: Theory and practical guidelines

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Plausible reasoning



Reasoning under uncertainty

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...with **probability**

What is probability, really?

Probability in antiquity

“That is probable which for the most part usually comes to pass, or which is a part of the ordinary beliefs of mankind, or which contains in itself some resemblance to these qualities, whether such resemblance be true or false.”

– Cicero, ca. 80 BCE

The classical formulation

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Doesn't generalize; what if not all outcomes are equally likely?

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- Based on circular reasoning

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How can such probabilities be verified? How can we know they are *useful*?

Probability as logic

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$\bar{A}$ is the logical *complement* (NOT)

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Generalized sum rule

$$\Pr(A + B) = \Pr(A) + \Pr(B) - \Pr(A, B)$$

Bayes' theorem

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A tale of two statisticians

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For the Bayesian, frequency-based probability is a special case

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Answer: We don't have enough information!

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$$\Pr(\text{sick} \mid \text{positive}) = \frac{99}{990/2 + 99} \approx 0.17 \times 100 = 17\%$$

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$\mathcal{H}$	$\Pr(\mathcal{H})$	$\Pr(\text{positive} \mid \mathcal{H})$	$\Pr(\text{positive} \mid \mathcal{H})\Pr(\mathcal{H})$
sick	0.01	0.99	0.0099
not sick	0.99	0.05	0.0495

$$\begin{aligned}\Pr(\text{sick} \mid \text{positive}) &= \frac{\Pr(\text{positive} \mid \text{sick})\Pr(\text{sick})}{\Pr(\text{positive})} \\ &= \frac{0.0099}{0.0099 + 0.0495} \approx 0.17\end{aligned}$$

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Failing to account for $\Pr(\text{sick})$ is called **base rate neglect**

The more general case

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But the Frequentist would object that β cannot have a probability

Null hypothesis significance testing

1. Define your experiment, along with the relevant **null** ($\mathcal{H}_0$) and (implicit) **alternative** ($\mathcal{H}_A$) hypotheses
2. Collect your data
3. Compute the relevant *test statistic* $\hat{s}$ from the data
4. Compute the so-called “p-value” $p = \Pr(s > \hat{s} \mid \mathcal{H}_0)$, i.e. the probability of observing a statistic *more extreme* than what you saw
5. Reject $\mathcal{H}_0$ if $p < \alpha$ where α is an arbitrary “significance” level.

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 - It says nothing about the alternative hypothesis

Problems with NHST

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- When is the “true” linear relationship between two values ever exactly zero?
- When is the “true” difference between two groups every exactly zero?
- Why should we think that such simple models are “true” in the first place?

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This means that, as $n \rightarrow \infty$, **all effects become significant**

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“What the use of p -values implies, therefore, is that a hypothesis that may be true may be rejected because it has not predicted observable results that have not occurred. This seems a remarkable procedure.”

– Harold Jeffreys (1961, p. 385)

4. Statistical test procedures make **non-obvious assumptions**

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Student's t-test assumptions

- The sample mean is normally distributed
- The sample variance is χ^2 distributed (or the samples are normally distributed)
- The sample mean and sample variance are independent
- The variances of both populations are equal (relaxed for Welch's t-test)

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Unlikely data under the null is not evidence for an alternative

What about confidence intervals?

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1. A particular 95% confidence interval **does not** have a 95% probability of containing the true parameter value
2. Smaller confidence intervals **do not** imply "more precise" results
3. Confidence intervals **do not** necessarily correspond to plausible or likely parameter values

Criticisms of NHST

These criticisms are not new...

Berkson J. Some difficulties of interpretation encountered in the application of the chi-square test. Journal of the American Statistical Association. **1938.**

Rozeboom WW. The fallacy of the null-hypothesis significance test. Psychological bulletin. **1960.**

Berger JO, Sellke T. Testing a point null hypothesis: The irreconcilability of p values and evidence. Journal of the American statistical Association. **1987.**

Johnson DH. The insignificance of statistical significance testing. The journal of wildlife management. **1999.**

Criticisms of NHST

...and have continued over time

Gelman A, Stern H. The difference between “significant” and “not significant” is not itself statistically significant. The American Statistician. **2006**.

Vasishth S, Mertzen D, Jäger LA, Gelman A. The statistical significance filter leads to overoptimistic expectations of replicability. Journal of Memory and Language. **2018**.

Wasserstein RL, Schirm AL, Lazar NA. Moving to a world beyond “ $p < 0.05$ ”. **2019**.

Amrhein V, Greenland S, McShane B. Scientists rise up against statistical significance. Nature. **2019**.

Remarks on correlation analysis

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- Are any of the variables **mathematically related** *a priori* (e.g., percentages)?

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Lack of quality control in science (see recent replication crises)

Recommendations: The Don'ts

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Do not selectively show results based on p-values or confidence intervals

Do not highlight certain results as “statistically significant” based on their p-values or confidence intervals

Do not apply “cookie-cutter” statistical analyses without thinking carefully about assumptions

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Do use resampling or out-of-sample testing to assess robustness of your results

Do prefer Neyman-Pearson hypothesis testing or Bayesian analyses over NHST

When to still be frequentist

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Nonparametric methods for large, high-dimensional datasets

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Models of interest are complex, nonlinear, or hierarchical

When to be nothing

If your experiment needs statistics, you ought to have done a better experiment.

– Ernest Rutheford (allegedly)

Bayesian inference

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Recall that probability distributions encode **uncertainty**

- **Unobserved variables** → Epistemic uncertainty (limited information)
- **Observed variables** → Aleatoric uncertainty (unexplained variability)

Building generative models

Suppose we want to model soil temperature

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$$T_{\text{soil}} \sim \mathcal{N}(\mu = \bar{T}_{\text{soil}} + \beta T_{\text{air}}, \Sigma = \varepsilon^2 I)$$

$$\bar{T}_{\text{soil}} \sim \mathcal{N}(\mu = 0, \sigma^2 = 100)$$

$$\beta \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

$$\log \varepsilon \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

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We could make this model *fully generative* by also modeling T_{air}

Building generative models

Suppose we want to model soil temperature

$$T_{\text{soil}} \sim \mathcal{N}(\mu = \bar{T}_{\text{soil}} + \beta T_{\text{air}}, \Sigma = \varepsilon^2 I)$$

$$\bar{T}_{\text{soil}} \sim \mathcal{N}(\mu = 0, \sigma^2 = 100)$$

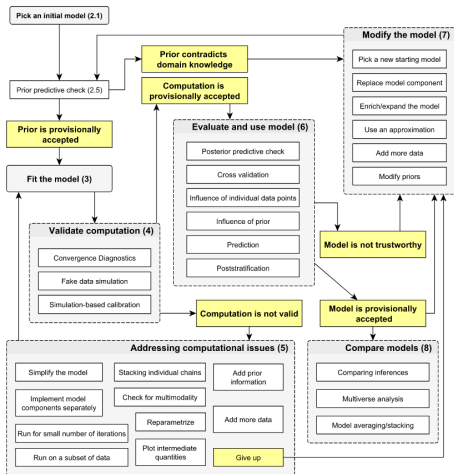
$$\beta \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

$$\log \varepsilon \sim \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

The β parameter relates T_{soil} and T_{air}

We could make this model *fully generative* by also modeling T_{air}

The Bayesian Workflow



The Bayesian Workflow (Gelman et al. 2020)

Bayesian inference

Bayes' rule gives a way to condition the **unobserved** variables in the model on the **observed** variables (data):

$$p(\theta \mid \mathbf{y}, \mathbf{x}) \propto p(\mathbf{y} \mid \theta, \mathbf{x}) p(\theta \mid \mathbf{x})$$

In the previous example, $\theta = [\bar{T}_{\text{soil}}, \beta, \varepsilon]$, $\mathbf{y} = T_{\text{soil}}$, and $\mathbf{x} = T_{\text{air}}$

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Once we have a summary of $p(\theta \mid \mathbf{y}, \mathbf{x})$, we can ask straightforward questions like:

What is the probability that the linear effect (slope) $\beta > 0$?

Maximum likelihood estimation (MLE)

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Typically, this is done via the negative log likelihood:

$$\mathcal{L}(\theta) = -\log p(\mathbf{y} \mid \theta)$$

and solving $\frac{\partial \mathcal{L}}{\partial \theta} = 0$ for θ .

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Common loss functions you may have used:

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All are forms of maximum likelihood

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Maximum a posteriori (MAP) estimation maximizes $p(\theta | \mathbf{y}) \propto p(\mathbf{y} | \theta)p(\theta)$ with respect to θ .

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Notice that maximum likelihood estimation (MLE) is just a special case of MAP.

Estimating the posterior distribution

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The goal is evaluate **expectations** of the form:

$$\mathbb{E}[f(\theta)] = \int_{\Theta} f(\theta)p(\theta \mid \mathbf{y})d\theta$$

Unfortunately, that's pretty hard!

Markov Chain Monte Carlo

Enter... Markov Chain Monte Carlo! (MCMC)

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Modern state-of-the-art samplers are typically variants **Hamiltonian Monte Carlo** (HMC) which use gradient information

Hamiltonian Monte Carlo

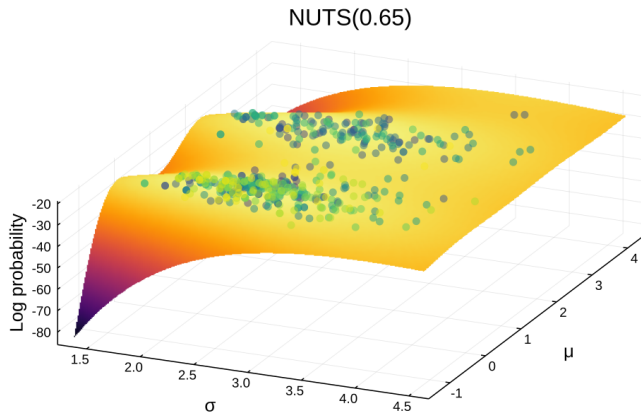


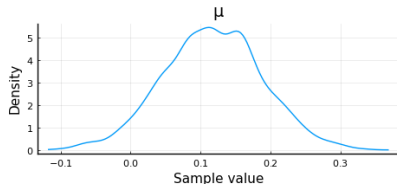
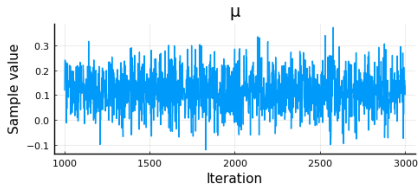
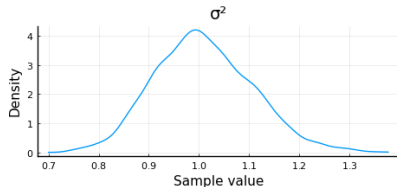
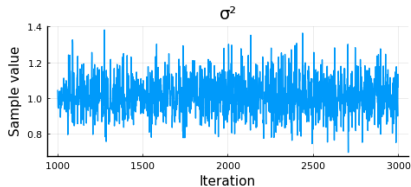
Figure 1: NUTS sampler visualization (source: Turing.jl documentation)

Interpreting the output

The output of MCMC and HMC algorithms is sequence of samples, typically referred to as a chain or a trace.

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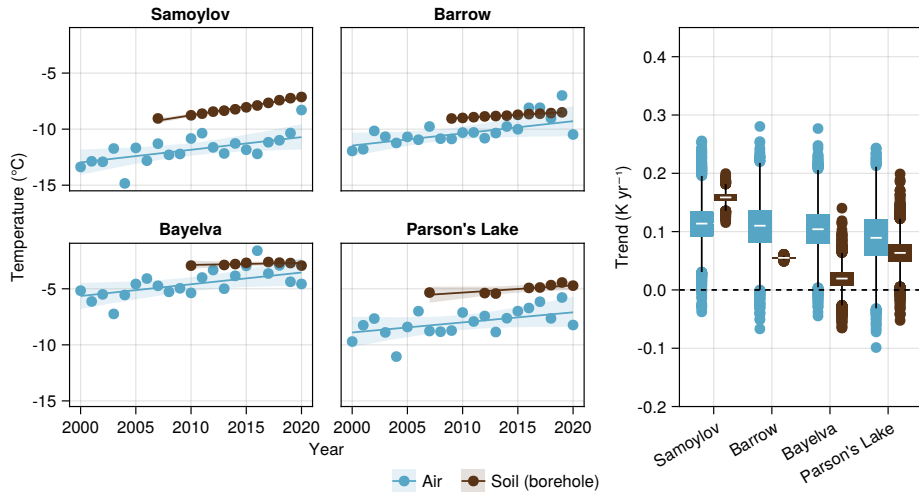
Overview of algorithms for Bayesian inference

- Markov Chain Monte Carlo (MCMC)
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- Ensemble Kalman Sampling (EKS)

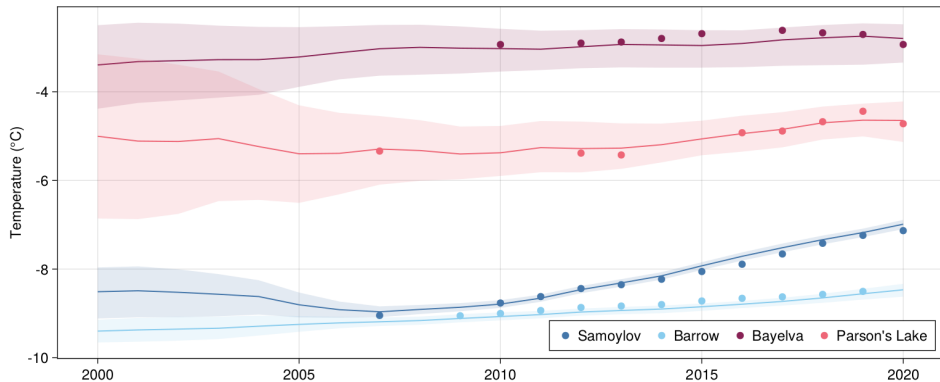
Overview of algorithms for Bayesian inference

	Forward runs	Gradients required	Parallelizable	Asymptotic convergence
MCMC	10^4 to 10^6	No	No	Yes
HMC	10^3 to 10^5	Yes	No	Yes
VI	10^2 to 10^4	Yes	Partially	No
SMC	10^3 to 10^5	No	Yes	Yes
EKS	10^3 to 10^4	No	Yes	Gaussian

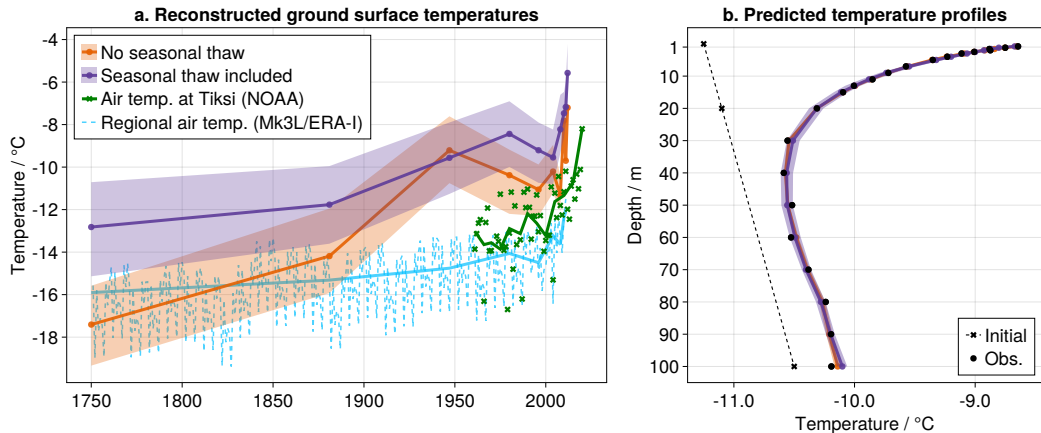
Use case: Multi-site trend analysis



Use case: Inverse modeling of heat transfer with EKS



Use case: Surface temperature reconstruction



Key limitations and challenges

Bayesian inference provides a flexible and intuitive framework for solving inverse problems and building generative models...

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- Software support

Software tools for Bayesian inference

Modern Bayesian workflows are implemented using **probabilistic programming languages** (PPLs)

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Python

- PyMC and bambi
- pyro (pytorch)
- numpyro (JAX)
- arviz (plotting)

R

- rstan and brms
- tidybayes
(plotting)

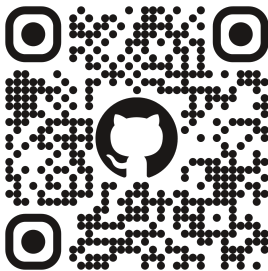
Julia

- Turing.jl
- Gen.jl
- ArviZ.jl
(plotting)

Part II: Bayes by example

Follow along!

Part II of the workshop consists of a series of Jupyter notebooks.



You can directly open these notebooks on Google Colab if you do not want to bother setting up a new python environment!